# Lecture 1: Overfitting, Hypothesis Testing, and Multiple Testing

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Regarding the Kaggle problem

## 1 Example 1: Public vs. Private Test Sets, Overfitting, and a Hypergeometric Model

### 1.1 Setup and notation

- Two disjoint test sets:

$$n_{\text{public}},\ n_{\text{private}}, \qquad n_{\text{total}} := n_{\text{public}} + n_{\text{private}}.$$

- Binary outcome per test item: correct/incorrect, encoded as 1/0.
- For a given team/model, let

$$K_{\text{total}} := \#\text{ correct on the entire (public + private) test set,}$$

and

$$K_{\text{public}} := \#\text{ correct on the public set,} \qquad K_{\text{private}} := K_{\text{total}} - K_{\text{public}}.$$

### 1.2 Key modeling assumptions (what must be true for the test below)

The hypergeometric model below is appropriate if:

1. **No leakage / no adaptive dependence on the split:** the model is effectively evaluated on the split without having tuned to the public subset (in practice, this is the fragile assumption of interest).
2. **Exchangeability of items across the split:** conditional on $K_{\text{total}}$, which items are correct is "as if random" with respect to whether an item landed in the public or private set. A sufficient condition is that the public set is a simple random sample from the total test set (or both are i.i.d. from the same population and the split is random).

If teams repeatedly submit and adapt to public feedback, (1) is typically violated.

### 1.3 Hypergeometric distribution for $K_{\text{public}}$ (conditional on $K_{\text{total}}$)

Condition on the realized overall performance $K_{\text{total}}$. Think of the full test set as an urn with:

$$N = n_{\text{total}} \quad \text{items total}, \qquad K = K_{\text{total}} \quad \text{“success” items (correct)},$$

and we “draw”

$$n = n_{\text{public}} \quad \text{items into the public set.}$$

Then under the assumptions above,

$$K_{\text{public}} \sim \text{Hypergeometric}(N, K, n),$$

i.e.

$$\Pr(K_{\text{public}} = k \mid K_{\text{total}}) = \frac{\binom{K_{\text{total}}}{k}\binom{n_{\text{total}}-K_{\text{total}}}{n_{\text{public}}-k}}{\binom{n_{\text{total}}}{n_{\text{public}}}},$$

for integers $k$ in the feasible range

$$\max\{0,\, n_{\text{public}} - (n_{\text{total}} - K_{\text{total}})\} \le k \le \min\{n_{\text{public}},\, K_{\text{total}}\}.$$

The reason it’s proper is because the public set is formed without replacement (no independent trials, no success probability p)

### 1.4 A one-sided p-value for suspiciously good public performance

If the concern is **overfitting to the public set**, then we look for unusually large $K_{\text{public}}$ relative to what would be expected under the null model above. A natural one-sided p-value is the right tail:

$$p = \Pr\bigl(K_{\text{public}} \ge k_{\text{obs}} \mid K_{\text{total}}\bigr) = \sum_{k=k_{\text{obs}}}^{\min(n_{\text{public}},K_{\text{total}})} \Pr(K_{\text{public}} = k \mid K_{\text{total}}).$$

Interpretation: small $p$ suggests the public score is too high to be explained by random splitting alone (given the overall total correct).

### 1.5 Notes / open issues to explore

- **Shared randomness across teams:** The public/private split is the same for all teams, so $K_{\text{public}}$ values are not independent across teams. This matters if you are making *many* such inferences across teams (multiple testing / selection).
- **Repeated submissions and adaptivity:** If teams tune based on public feedback, then conditioning only on $K_{\text{total}}$ may not remove the dependence between correctness and being in the public set. This can bias the hypergeometric p-values (typically making public performance look “too good” more often than the null predicts).
  - Why the logic breaks under repeated submissions: The key assumption is exchangeability. But if repeatedly submitted, the model selectively improves on public items and correctness is no longer independent of being public
  - This is the same phenomenon as: optional stopping; data snooping; p-hacking
- **Selection effects:** Leaderboards usually highlight top performers; even under a valid null, the top-ranked teams can look extreme due to selection (again: multiple testing/selection).

# 2 Example 2: Hypothesis Testing and Multiple Testing (first 3 weeks)

## 2.1 Framework for hypothesis testing (HT)

A standard HT setup consists of:

1. **Null hypothesis $H_0$:** a baseline claim (often "no effect"), typically something we seek evidence against.
2. **Test statistic $T$:** a function of the data used to quantify evidence against $H_0$.
3. **Null distribution of $T$:** the distribution of $T$ assuming $H_0$ is true.
4. **Decision rule (a test):** a map from the observed $T$ to *reject* $H_0$ or *do not reject* $H_0$.

## 2.2 Type I error and level

A test has **Type I error level** $\alpha$ (is a **level-$\alpha$ test**) if

$$\Pr_{H_0}(\text{reject } H_0) \leq \alpha.$$

## 2.3 p-values (correct definition and key property)

A **p-value** is a data-dependent quantity $p \in [0, 1]$ such that, under $H_0$,

$$\Pr_{H_0}(p \leq t) \leq t \qquad \text{for all } t \in [0, 1].$$

This is the **super-uniform** property. In many continuous, correctly-specified settings, p-values are exactly uniform under $H_0$ (i.e. equality holds), but *super-uniform* is the general requirement.

- Under the null, p-values do not pile up near 0
- Small p-values are genuinely rare if $H_0$ is true

Interpretation: the p-value is the *smallest* significance level $\alpha$ at which the observed data would lead to rejection, under the chosen test.

## 2.4 Multiple testing / multiple comparisons

We test $m$ null hypotheses:

$$H_{0,1},\ H_{0,2},\ \ldots,\ H_{0,m},$$

with corresponding p-values $p_1, \ldots, p_m$.

The goal is to choose a set of rejections while controlling false positives under some criterion.

## 2.5 Common false-positive control goals

Let $V$ be the number of false rejections and $R$ the total number of rejections. Common targets include:

- **FWER:** control $\Pr(V \geq 1)$ (probability of *any* false rejection).

- **k-FWER / tail control:** control $\Pr(V \geq k)$ for some $k \geq 1$ (e.g. "more than $x$% false" variants).
- **Expected counts:** control $\mathbb{E}[V]$.
- **FDR:** control $\mathbb{E}\left[\frac{V}{\max(R,1)}\right]$ (expected false discovery proportion).

### 2.6 Global null and what p-values look like

Define the **global null**:

$$H_{0,\text{global}} := \bigcap_{i=1}^{m} H_{0,i}.$$

- If $H_{0,\text{global}}$ is true and p-values are continuous and valid, then each $p_i$ is (super-)uniform, and the collection $\{p_i\}$ typically looks roughly uniform overall (up to dependence and discreteness). (this is a detection problem now, not identification problem)
- If $H_{0,\text{global}}$ is false, departures from uniformity can look like:
  1. **Sparse strong signals:** most nulls true, a small number false with strong effects $\Rightarrow$ a few extremely small p-values.
  2. **Dense weak signals:** many nulls false but effects are weak $\Rightarrow$ p-values are generally shifted smaller than uniform (more mass near 0, but not necessarily extreme).
  3. **All nulls are true:** Roughly uniform p-values

### 2.7 Testing the global null: Bonferroni test

A simple level-$\alpha$ test of $H_{0,\text{global}}$ is the Bonferroni/min-p test:

$$\text{Reject } H_{0,\text{global}} \quad \text{if} \quad \min_{1 \leq i \leq m} p_i \leq \frac{\alpha}{m}.$$

Justification (no independence needed): by the union bound and the super-uniform property,

$$\Pr_{H_{0,\text{global}}}\left(\min_i p_i \leq \frac{\alpha}{m}\right) = \Pr_{H_{0,\text{global}}}\left(\bigcup_{i=1}^{m}\left\{p_i \leq \frac{\alpha}{m}\right\}\right) \leq \sum_{i=1}^{m} \Pr\left(p_i \leq \frac{\alpha}{m}\right) \leq \sum_{i=1}^{m} \frac{\alpha}{m} = \alpha.$$

### 2.8 Note (still to address later)

Bonferroni is conservative when tests are positively correlated or when $m$ is large; more powerful procedures (e.g. Holm, Hochberg, BH) depend on the chosen error criterion and assumptions.